

RetailMart Sales Analytics

EXECUTIVE POWER BI DASHBOARD REQUIREMENTS DOCUMENT (DRD)

Project: Retail Sales Performance Analytics **Prepared By:** Senior AI Business Analyst **Target Audience:** Executive Leadership Board

1. Executive Summary

RetailMart requires an enterprise-grade, interactive Power BI executive dashboard designed to provide real-time visibility into comprehensive sales performance, profitability trends, customer behaviors, product efficiency, and regional dynamics. Currently, leadership relies on fragmented reporting mechanisms and delayed manual spreadsheets, hindering rapid strategic decision-making.

This Dashboard Requirements Document (DRD) defines the complete functional and technical specification for the RetailMart Power BI solution. The dashboard will integrate core transactional, customer, and supply chain data sources into a unified analytical model, empowering C-suite executives and regional directors to monitor key performance indicators (KPIs), drill down into operational bottlenecks, and execute data-driven interventions.

2. Dashboard Objectives

- The primary objectives of the RetailMart Executive Dashboard are to:
- **Centralize Enterprise Visibility:** Provide a single source of truth for revenue, profit, and operational metrics across all retail channels and geographic regions.
 - **Enable Proactive Performance Management:** Empower leadership to track performance against strategic targets and rapidly identify underperforming stores, categories, or customer segments.
 - **Accelerate Decision-Making:** Eliminate manual data gathering cycles by delivering automated, near-real-time updates and intuitive drill-through capabilities.
 - **Optimize Profitability:** Highlight margin leakage, markdown inefficiencies, and inventory holding costs to guide corrective pricing and assortment strategies.

3. Business Questions

The dashboard is structured to answer core strategic questions posed by executive management:	
ANALYTICAL DOMAIN	CORE BUSINESS QUESTION
Regional Performance	Which geographic region and store locations generate the highest revenue and net profit margins?
Product Profitability	Which specific products and sub-categories yield the highest profit margins versus those diluting overall profitability?
Customer Segmentation	Which customer segments (e.g., Consumer, Corporate, Home Office) contribute the highest sales volume and repeat purchase rates?
Category Analysis	Which product categories underperform against quarterly sales targets and require promotional intervention?
Sales Trends	What is the monthly and seasonal sales trajectory compared to historical baselines and annual forecasts?

4. Dashboard Pages Structure

The Power BI report is structured into six dedicated analytical pages tailored to different executive personas:	
DASHBOARD PAGE	PAGE PURPOSE & TARGET PERSONA
1. Executive Overview	High-level summary for C-suite leadership displaying enterprise KPIs, monthly sales trends, and regional heatmaps at a glance.
2. Sales Performance	Detailed revenue breakdown by channel, store type, and sales rep, tracking performance against monthly and quarterly quotas.
3. Profit Analysis	Granular margin analysis examining cost of goods sold (COGS), discounting impact, and net profit contributions by category.
4. Customer Analysis	Segmentation metrics analyzing customer acquisition, retention rates, lifetime value (CLV), and purchasing frequency.
5. Product Analysis	SKU-level performance tracking inventory turnover, top-selling items, dead stock, and category contributions.
6. Regional Analysis	Geographic drill-down evaluating store-level productivity, regional sales density, and localized market share.

5. Core KPI Cards

Each dashboard page prominently features standard executive KPI cards with conditional formatting and YoY variance indicators:

KPI METRIC	CALCULATION LOGIC / FORMULA	BUSINESS PURPOSE
Total Sales	$\sum (\text{Quantity} \times \text{Unit Price}) - \text{Discounts}$	Measures total gross revenue velocity.
Total Profit	$\sum (\text{Total Sales} - \text{Total COGS})$	Tracks absolute financial earnings after expenses.
Profit Margin	$(\text{Total Profit} / \text{Total Sales}) \times 100$	Evaluates pricing efficiency and operational profitability.
Orders	$\text{Distinct Count}(\text{Order ID})$	Measures transaction volume and customer checkout activity.
Average Order Value (AOV)	$\text{Total Sales} / \text{Total Orders}$	Tracks basket size and upselling effectiveness.
Customer Count	$\text{Distinct Count}(\text{CustomerID})$	Monitors active buyer base and customer reach.
Sales Growth	$((\text{Sales}_{\text{Current}} - \text{Sales}_{\text{Prior}}) / \text{Sales}_{\text{Prior}}) \times 100$	Tracks YoY or MoM top-line expansion trajectory.

6. Visualizations & Chart Specifications

CHART NAME	PURPOSE	BUSINESS VALUE	CHART TYPE
Monthly Sales Trend	Display revenue trajectory over time with target comparison.	Identifies seasonal peaks, sales slumps, and annual growth velocity.	Line & Column Chart
Regional Revenue Map	Geographic bubble map displaying sales density by state/city.	Highlights top-performing regions and underperforming territories.	Filled / Bubble Map
Category Profitability	Compare revenue versus profit margin across product categories.	Exposes high-revenue low-margin categories needing pricing review.	Clustered Bar Chart
Customer Segment Share	Show sales contribution percentage by customer segment.	Guides targeted marketing allocation toward high-value cohorts.	Donut Chart
Top 10 Products Matrix	List top-performing SKUs by sales, profit, and units sold.	Informs inventory replenishment and merchandising priorities.	Table / Matrix
Discount vs. Margin Scatter	Analyze correlation between discount depth and net profit margin.	Prevents margin erosion from uncoordinated promotional markdowns.	Scatter Plot

7. Global & Page-Level Filters (Slicers)

To enable interactive data slicing, the dashboard incorporates a standardized filter panel across all pages:

Filter Name	Filter Behavior & Scope
Year & Month	Global time slicer supporting multi-select and relative date filtering (e.g., Year-to-Date, Last 12 Months).
Region & State	Hierarchical geographic slicer allowing cascade filtering from regional hubs down to individual states.
Store	Specific store selector allowing individual retail location performance audits.
Category & Sub Category	Product taxonomy filter to isolate performance within specific merchandise lines.
Customer Segment	Demographic filter (Consumer, Corporate, Home Office) for targeted behavioral analysis.
Payment Method	Transaction attribute filter (Credit Card, Cash, Digital Wallet, Financing).

8. Drill-Through & 9. Navigation Architecture

8. Drill-Through Functionality

Users can right-click any data point on regional, store, or product visuals to drill through to dedicated detail pages. For example, clicking a specific store on the regional map dynamically passes the store context to a granular **Store Performance Drill-Through Page** displaying employee sales metrics, local inventory turnover, and daily foot traffic.

9. Navigation Architecture

The dashboard utilizes a clean, modern sidebar navigation pane featuring bookmark-driven page switching and SVG icons. A persistent breadcrumb header and home button ensure intuitive user flow, while conditional button states highlight the active page.

10. Acceptance Criteria

The Power BI dashboard will be formally signed off upon meeting the following criteria:

- **Data Accuracy:** Total sales and profit figures reconcile with the enterprise ERP and financial ledger within a 0.01% tolerance threshold.
- **Performance Benchmark:** All report pages and visuals render completely within 3.0 seconds upon initial load or slicer selection.
- **Row-Level Security (RLS):** Regional managers successfully authenticate and view only data corresponding to their assigned geographic territory.
- **Mobile Responsiveness:** Layouts are optimized for Power BI Mobile view, ensuring seamless readability on executive tablets and smartphones.

11. Future Dashboard Enhancements

Phase 2 dashboard roadmap items scheduled for subsequent release include:

- **AI Insights & Forecasting:** Integration of Power BI Decomposition Trees and ML-powered sales forecasting visuals predicting next-quarter revenue.
- **Customer Churn Predictive Scoring:** Inclusion of machine learning badges highlighting customers at high risk of churn.
- **Automated Executive Alerts:** Integration with Microsoft Teams to trigger automated alert notifications when store margins or inventory stockouts breach critical thresholds.
- ***External Market Benchmarking:** Overlaying regional macroeconomic indices and competitor foot-traffic data for external market share context.